

Up-Regulation of the Uterine Estrogen Receptor and Its Messenger Ribonucleic Acid during the Mouse Estrous Cycle: The Role of Estradiol*

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ABSTRACT. Uterine estrogen receptor (ER) and ER mRNA were measured in cycling and ovariectomized (OVX) estrogen-treated mice to probe the physiological regulation of the intracellular distribution and biosynthesis of ER. On proestrus, when plasma estradiol (E_2) levels are highest, the cell nuclear ER concentration was 2.4-fold greater than on metestrus. This increase was primarily attributable to an increase in total cellular ER (cytosolic plus nuclear ER) and only secondarily to an activation of ER, as measured by its redistribution from the cytosolic (i.e. nuclear-extractable) to the nuclear (nonextractable) fraction. Total cellular ER concentration was 1.8-fold higher on proestrus than on metestrus, whereas the fraction of total ER in the nuclear compartment (i.e. the percentage activated) was only 1.3-fold higher. The concentration of cellular ER mRNA was 3-fold greater on proestrus than on the other days

of the estrous cycle, suggesting that the increased concentration of ER on proestrus was a consequence of increased ER gene expression. In OVX mice, physiological and, to a lesser extent, supraphysiological levels of E_2 increased cell nuclear ER. As in proestrous mice, the increased ER content contributed more than ER activation to the increased cell nuclear ER concentration. Physiological, but not supraphysiological, concentrations of E_2 increased ER mRNA in OVX mice. Together, these results suggest that up-regulation by E_2 of ER mRNA and ER accounts for most of the increased nuclear binding of ER on proestrus. E_2 -dependent activation and consequent DNA binding of ER presumably initiate this process, but quantitatively account for only a small fraction of the increased nuclear binding of ER. (*Endocrinology* 130: 1923-1930, 1992)

ESTRADIOL (E_2) plays a major role in the regulation of reproductive function. During the estrous cycle, the preovulatory increase in plasma E_2 acts on the brain to promote female mating behavior (1, 2) and activate neuroendocrine mechanisms, leading to the preovulatory release of LH (3-5). In the uterus, E_2 induces a wide spectrum of cellular changes in preparation for the possibility of pregnancy, including endometrial hypertrophy and hyperplasia (6, 7) and the induction of progesterone receptors (8). E_2 effects these changes by interacting with the estrogen receptor (ER), a ligand-dependent transcriptional regulatory protein that alters gene expression by binding to cognate DNA sequences and

interacting with other transcriptional regulatory factors (9).

The structure of the ER and its interaction with estrogen-responsive genes are under intense study, and as part of these investigations, the effects of estrogen on the expression of its own receptor have been examined. However, most studies have focused on regulation of the ER gene in breast cancer cell lines (10-13), and relatively little attention has been given to the physiological regulation of ER *in vivo*.

During the estrous cycle, the concentration of ER in cell nuclei fluctuates in parallel with the pattern of plasma E_2 , rising to a maximal level on proestrus, just before ovulation (14, 15). Since increased nuclear binding of ER can be induced by E_2 through the formation of an activated E_2 -ER complex with higher affinity for DNA (16), the increased nuclear binding of ER on proestrus could arise from an increased activation of ER. However, other mechanisms may play a role. In the course of studying the physiological regulation of ER mRNA, we discovered and report here that a more important mechanism involves a dramatic increase in the intracellular

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concentration of ER mRNA and, as a consequence, ER protein available for nuclear binding. We hypothesized that the preovulatory rise of E₂ was responsible for the increases in ER and ER mRNA. Because earlier studies using supraphysiological concentrations of E₂ had reported that E₂ suppresses ER mRNA in the uterus (17) and brain (18, 19) of rodents, we hypothesized further that the effect of E₂ on ER and its mRNA is dose dependent; physiological concentrations of E₂ increase ER and ER mRNA levels, although supraphysiological concentrations may not. The results of the present study are consistent with these hypotheses.

Materials and Methods

Experimental design and nomenclature

Two experiments were conducted. In the first experiment, the profiles of uterine nuclear and cytosolic ER and ER mRNA concentrations during the estrous cycle were determined. The second experiment was designed to determine the role of E₂ in shaping the profiles of ER and ER mRNA that were observed during the estrous cycle, and the effects of physiological *vs.* supraphysiological concentrations of E₂ on ER biosynthesis.

In this report nuclear ER refers to receptors retained in the nuclear fraction after cell fractionation, presumably because they are tightly bound to chromatin. Cytosolic ER consists of receptors in the cytosol, presumably, because although nuclear *in vivo*, they are only loosely bound to chromatin and are solubilized during cell fractionation.

Animals and tissue collection

Virgin female C57BL/6J mice were obtained from the Jackson Laboratory (Bar Harbor, ME) at 5 weeks of age, housed three per cage, and maintained as described previously (20). Animals in this study were 3–6 months old. In the first experiment, vaginal smears were collected daily and classified according to the stage of the estrous cycle (21). Only animals with at least two consecutive cycles of 5 days or less before death were included. Animal husbandry and experimentation were conducted in accord with the principles and procedures outlined in the Guidelines for Care and Use of Experimental Animals.

Mice were anesthetized with 2,2,2-tribromoethanol (BDH Chemicals, St. Laurent, Quebec, Canada) (22) and exsanguinated from the right ventricle. Plasma was frozen (–20 C) for E₂ RIA (23). For studies of ER mRNA, uteri were dissected and stored at –80 C until assayed. Measurements of ER binding were made on freshly dissected uteri. Uteri were weighed, and a small weighed portion from each uterus was saved for DNA determination by the diphenylamine assay (24).

Reagents

[³²P]CTP (800 Ci/mmol) was obtained from New England Nuclear (Boston, MA). Restriction enzymes were purchased from IBI (New Haven, CT) and New England Biolabs (Beverly, MA). RNase-A (type 1-A), RNase-T1 (grade IV), and protease-K (type XI) were purchased from Sigma (St.

Louis, MO). The mouse ER (MER) cDNA clone was kindly provided by Dr. M. Parker, Imperial Cancer Research Fund (London, United Kingdom) (25). The vectors used for subcloning (pGEM) and all reagents required for cRNA synthesis were obtained from Promega Biotec (Madison, WI). All other reagents were of molecular biology grade.

cRNA synthesis

The solution hybridization/RNase protection assay that was used to quantitate ER mRNA levels in this study is optimal when cRNA probes are less than 600 basepairs in length. Therefore, a 530 basepair restriction fragment of the MER cDNA encoding the A and B domains of the MER cDNA was subcloned into the polylinker sequence of a pGEM vector containing the transcriptional promoter for T7 RNA polymerase. ER mRNA content was determined using a radiolabeled antisense cRNA probe synthesized from this subclone (26). The integrity of the cRNA probe was verified on a 4% polyacrylamide gel to ensure that full-length transcripts were generated for hybridization.

RNA isolation and solution hybridization

Total RNA was isolated from frozen tissue by homogenization in guanidine isothiocyanate buffer, followed by centrifugation over cesium chloride (27). The RNA yield of each tissue was determined by UV spectrophotometry, assuming 1 OD₂₆₀ U = 40 µg/ml. Purified RNA pellets were resuspended in sterile water and stored at –80 C. The quality of the RNA was monitored by 1.0% agarose formaldehyde gel electrophoresis.

The solution hybridizations were performed as previously described (28) with small modifications. Briefly, 26 µl of the ³²P-labeled cRNA probe were resuspended in hybridization buffer (40 mM PIPES, pH 6.7; 400 mM NaCl; and 1 mM EDTA, pH 7.5, in 80% formamide) at 10,000 cpm/µl and mixed with 4 µg total uterine RNA in 4 µl water. The samples were heated for 5 min at 90 C, and then kept at 55 C overnight (16–18 h). After hybridization, the samples were chilled on ice for 5 min, and 10 vol digestion buffer (300 mM NaCl; 10 mM Tris, pH 8.0; and 5 mM EDTA, pH 8.0) containing RNase-A (40 µg/ml) and RNase-T1 (2 µg/ml) were added to degrade unhybridized single stranded RNA. After a 30-min incubation at 25 C, the reactions were terminated by the addition of 20 µl 10% sodium dodecyl sulfate (SDS) and 1 µl protease-K (10 mg/ml) for 15 min at 37 C. The mixture was extracted with phenol-chloroform-isoamylalcohol (25:24:1) and then with chloroform-isoamylalcohol alone. The aqueous phase was removed, 50 µg yeast tRNA were added as carrier, and the samples were precipitated with 0.2 vol ammonium acetate and 2.5 vol 100% ethanol for 15 min at –70 C, and 15 min at –20 C. The pellets were washed with 70% ethanol, dried, and resuspended in 20 µl gel-loading buffer [70% TBE (90 mM Tris-base, 90 mM boric acid, and 2.5 mM EDTA) and 30% dye mix [0.2% (wt/vol) bromophenol blue and 0.2% (wt/vol) xylene cyanol]]. The samples were heated twice at 65 C for 5 min, loaded onto a 5% nondenaturing polyacrylamide gel, and electrophoresed for 3.0–3.5 h at 290 V. The gels were dried and exposed to x-ray film (X-Omat, Eastman Kodak, Rochester, NY) overnight with one intensifying screen. A single protected band corresponding in length to the fully protected fragment was observed, as shown in Fig. 1. No detectable band

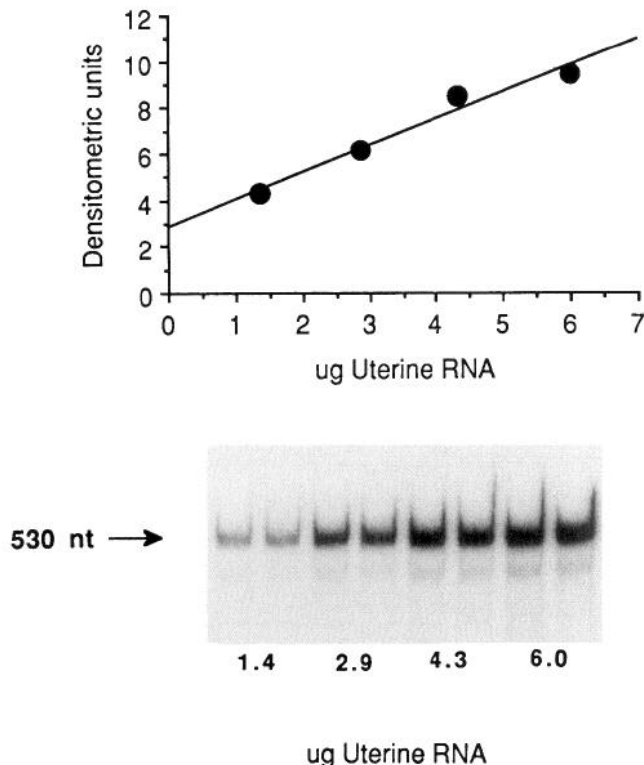


FIG. 1. Representative autoradiogram and resultant standard curve of serial dilutions, each in duplicate, of total uterine RNA hybridized to a ³²P-labeled 530-nucleotide (nt) antisense ER cRNA probe. Hybrids were measured using a solution hybridization/RNase protection protocol and analyzed by polyacrylamide gel electrophoresis, as described in *Materials and Methods*.

was seen after hybridization with yeast RNA, and a very low signal was observed after hybridization to RNA from brain cortex, in which ER is detectable but in very low abundance (our unpublished observations).

The optical densities of the protected bands of the experimental samples were measured with a Biorad Densitometer and compared to those of a four-point standard curve constructed from serial dilutions from a pool of uterine RNA that was hybridized with probe in each experiment (Fig. 1). The standard curves were linear ($r > 0.98$). All test samples and each concentration of the standard curve were run in duplicate. Because the pool of uterine RNA used as a standard in the measurement of ER mRNA during the estrous cycle differed from the pool used to measure ER mRNA after E₂ treatment, the densitometric units in the two experiments are not directly comparable.

Measurement of polyadenylated RNA

Polyadenylated RNA was measured by slot blot hybridization of uterine RNA with radiolabeled oligo-dT. Each uterine RNA sample was blotted in triplicate (1 μ g total RNA/replicate). Samples were brought to 120 μ l with H₂O, diluted with 150 μ l deionized formamide, heated for 5 min at 65 C, chilled on ice, and diluted with 0.1 vol 20 $\times$ SSC. An 80- μ l aliquot was applied to GeneScreen presoaked in 20 $\times$ SSC, using a Schleicher and Schuell slot blot manifold (Keene, NH). A six-

point standard curve of the hybridization signal constructed from serial dilutions of mouse liver RNA was blotted in triplicate in parallel with the test samples. RNA was UV cross-linked to the membrane in a Stratalinker (Stratagene, La Jolla, CA). Radiolabeled oligo-dT was synthesized using oligo-dT (Collaborative Research, Bedford, MA), terminal deoxynucleotidyl transferase, enzyme buffer (Bethesda Research Laboratories, Gaithersburg, MD), and [³²P]dTTP (New England Nuclear).

Filters were prehybridized overnight (16–18 h) at 42 C in 50% formamide; 750 mM NaCl; 20 mM Tris (pH 7.5); 1 $\times$ Denhardt's (200 μ g/ml each of BSA, polyvinyl pyrrolidone, and Ficoll); 200 μ g/ml total yeast RNA; 0.1% SDS; and 0.5% sodium pyrophosphate. Filters were then hybridized overnight at 37 C in a buffer of 6 $\times$ SSC, 20 mM sodium phosphate (pH 6.8), 0.1 mg/ml salmon sperm DNA, 1 $\times$ Denhardt's, 0.1 mg/ml total yeast RNA, 0.1% SDS, 0.2% sodium pyrophosphate, and 10% dextran sulfate containing 1.5 $\times$ 10⁶ cpm oligo-dT/1 ml buffer. After hybridization, filters were washed at room temperature for 20 min in 2 $\times$ SSC buffer, (0.2 $\times$ SSC, 0.1% SDS, 0.02 M sodium phosphate, and 0.05% sodium pyrophosphate) then for 1 h at 42 C in 0.2 $\times$ SSC wash buffer. Filters were exposed to x-ray film for 3 days with two intensifying screens. Autoradiographic signal intensities were quantified by densitometry and compared to the standard curve. Standard curves were linear and had correlation coefficients greater than 0.98.

ER measurements

Uterine ER was measured by NaSCN-potentiated exchange assay at 0 C validated for mouse ER (29). The receptor-hormone complex was separated from unbound hormone by hydroxylapatite adsorption chromatography (30). Specific binding in the cytosolic compartment is assumed to reflect nuclear ER *in vivo* that is solubilized from nuclei during cell fractionation because it is in an unactivated state with low affinity for DNA (16). The ER remaining in the nuclear compartment is presumed to be that tightly bound to chromatin or DNA and, hence, in an activated state (16). Total cellular ER is calculated as the sum of ER in the cytosolic and nuclear fractions.

Statistical analyses

The effects of stage of cycle and E₂ treatment on plasma E₂, ER mRNA, and poly(A) levels were assessed by one-way analysis of variance (ANOVA). The Student Newman-Keuls (SNK) multiple comparisons *a posteriori* test was used to determine the significance of differences between means. Effects of stage of cycle on ER levels were assessed by *t* test.

Results

ER and ER mRNA during the estrous cycle

Uteri and plasma were obtained from mice killed at midday on proestrus, estrus, metestrus, and diestrus. Cytosolic and nuclear ER were measured on proestrus and metestrus, when plasma E₂ is at its peak and nadir during the estrous cycle, respectively. ER mRNA was measured on all 4 days of the cycle.

Plasma concentrations of E₂ varied as expected during the estrous cycle ($P < 0.001$, by ANOVA; Table 1); they were highest on proestrus, fell to a nadir on estrus and metestrus, and began to rise again on diestrus. Although uterine weight also varied during the cycle, DNA content remained stable, indicating that there was little change in the cellular number (Table 1).

The concentrations of cell nuclear, cytosolic, and total (nuclear plus cytosolic) ER all differed markedly between proestrus and metestrus (Fig. 2). Nuclear ER was 2.4-fold higher on proestrus than on metestrus ($P < 0.001$). Increased nuclear binding of ER is generally believed to reflect an E₂-dependent transformation of ER into a species with greater affinity for DNA (16). As shown in Fig. 2, the fraction of ER in the nuclear compartment (and, hence, activated) only increased from 32% on metestrus to 41% on proestrus. This 1.3-fold increase accounts for only a fraction of the total increase in nuclear binding on proestrus. Indeed, most of the increase in nuclear ER is attributable to the increased total cellular ER content, which was 1.8-fold greater on proestrus than on metestrus and, thus, provided a larger pool of ER available for nuclear binding.

ER mRNA levels also varied during the estrous cycle (Fig. 3). ER mRNA expressed per μg uterine DNA or per whole uterus was 3-fold higher on proestrus than on the other days of the cycle ($P < 0.001$). The cellular concentrations of polyadenylated as well as total RNA were similarly increased on proestrus (Table 1), indicating that the increase in ER mRNA reflected a general increase in messenger and, presumably, ribosomal RNA content on proestrus. The ER mRNA concentration in ovariectomized (OVX) mice did not differ from basal values in intact mice (Fig. 3).

Effect of E₂ on ER and ER mRNA

To determine whether the increased ER and ER mRNA on proestrus could be attributable to the pre-ovulatory elevation of E₂, mice were OVX and, 2 days later, implanted for 2 days with capsules designed to produce physiological (4- and 8-mm long polyethylene

capsules) or supraphysiological (5-mm long Silastic capsules) concentrations of E₂. The two sizes of polyethylene capsules gave similar results and were, therefore, pooled. As shown in Table 2, E₂ levels in mice with polyethylene implants were not detectably higher than those in mice with blank implants. However, based on an earlier study (23), 4- and 8-mm implants would be expected to produce levels of E₂ 10 and 20 pg/ml higher than in OVX mice. The mice were clearly exposed to elevated levels of E₂, since their uterine weight was 3-fold greater than that of controls. Indeed, the uterine weight of mice exposed to these low levels of E₂ was greater than that of mice exposed to higher levels of E₂ ($P < 0.05$). The RIA used for this experiment (*cf.* Tables 1 and 2) had a higher background than that in the first experiment and was apparently unable to detect this increase. Mice exposed to Silastic implants had the expected supraphysiological plasma concentrations of E₂. The uterine weight of these mice was significantly higher than that of untreated OVX mice ($P < 0.05$), but, as noted, was significantly lower than that of mice exposed to the lower physiological concentrations of E₂.

Figure 4 shows the effects of the two concentrations of E₂ on ER content and intracellular distribution in OVX mice. Mice exposed to physiological levels of E₂ showed the greatest increase in total and nuclear ER. The nuclear ER level was 6-fold greater than that in OVX mice, and the total ER level was over 3-fold greater. Absolute concentrations of nuclear and total ER as well as the fractional binding of ER in the nuclear compartment were similar to those in mice on proestrus (*cf.* Figs. 2 and 4). Mice exposed to supraphysiological concentrations of E₂ also exhibited increased nuclear and total ER, although levels were less than those in mice exposed to physiological concentrations ($P < 0.05$). Figure 5 shows the effect of E₂ exposure on ER mRNA. Mice exposed to physiological concentrations of E₂ had 1.6-fold more uterine ER mRNA than OVX mice ($P < 0.05$). A similar trend was observed when ER mRNA was normalized to DNA, but the result did not quite reach statistical significance. Mice exposed to supraphysiological

TABLE 1. Uterine weight, DNA, RNA, and plasma concentrations of E₂ during the estrous cycle of C57BL/6J mice

	Estrus	Metestrus	Diestrus	Proestrus	P ^a
Plasma E ₂ (pg/ml)	4.2 $\pm$ 1.0 ^b	3.1 $\pm$ 0.7 ^b	7.5 $\pm$ 1.3 ^c	10.3 $\pm$ 1.1 ^c	<0.001
Uterine wt (mg)	78 $\pm$ 5 ^b	52 $\pm$ 3 ^c	61 $\pm$ 9 ^c	88 $\pm$ 4 ^b	<0.001
Total uterine DNA (μg)	648 $\pm$ 27	581 $\pm$ 75	545 $\pm$ 31	552 $\pm$ 96	NS
Total uterine RNA (μg)	207 $\pm$ 10 ^b	147 $\pm$ 11 ^c	132 $\pm$ 7 ^c	374 $\pm$ 24 ^d	<0.001
Total uterine Poly(A) RNA ^e	372 $\pm$ 37 ^b	308 $\pm$ 45 ^b	241 $\pm$ 46 ^b	723 $\pm$ 169 ^c	<0.01
n ^f	6–14	6–15	6–15	5–15	

Data shown are the mean $\pm$ SEM. Values with different alphabetic superscripts are significantly different ($P < 0.05$, by SNK).

^a Based on one-way ANOVA.

^e Densitometric units.

^f Sample size.

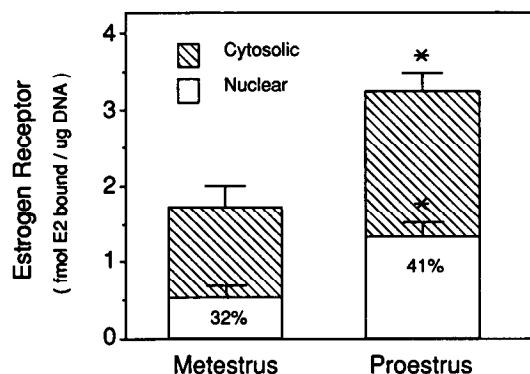


FIG. 2. Uterine ER content and its intracellular distribution on proestrus and metestrus. Percentages designate the fraction of total ER bound tightly to chromatin and defined as nuclear binding. Note that cytosolic refers to presumably loosely bound nuclear ER that is found in the cytosol as a result of cell fractionation procedures. As shown by *asterisks*, cytosolic and nuclear ER on proestrus were significantly different from corresponding values on metestrus.

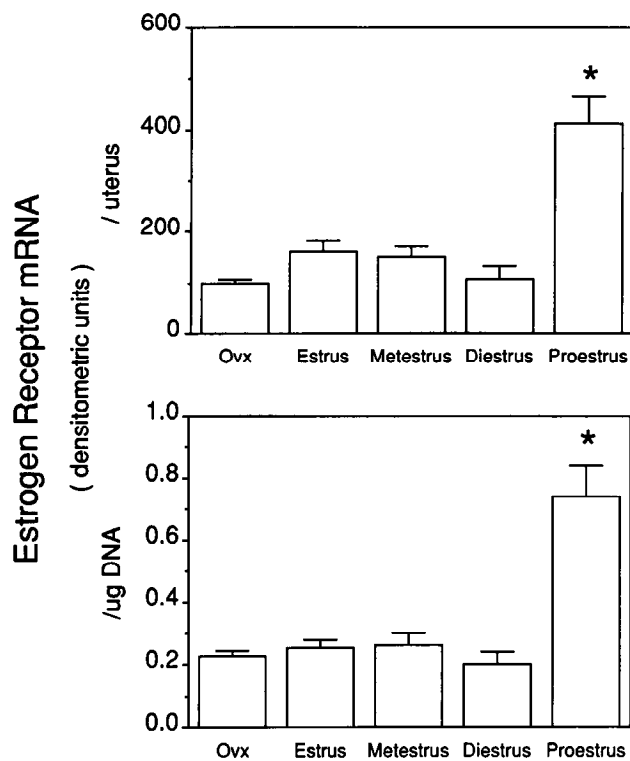


FIG. 3. Uterine ER mRNA content on proestrus, estrus, metestrus, diestrus, and 5 days after ovariectomy (OVX) in female mice. An *asterisk* indicates that the value is significantly different from all others ($P < 0.05$, by SNK).

ical concentrations of E₂ showed no increase in ER mRNA compared to OVX mice.

Table 2 shows the effect of E₂ treatment on uterine weight and DNA and RNA contents. Neither low nor high estrogen treatment influenced uterine DNA content. Although mice exposed to physiological concentrations of E₂ exhibited increased levels of total and polyadenylated RNA similar to those seen in mice on proes-

TABLE 2. Uterine weight, DNA, RNA, and plasma concentrations of E₂ in E₂-treated OVX mice

	Blank	Low E ₂	High E ₂	P ^a
Plasma E ₂ (pg/ml)	22 ± 2 ^b	24 ± 2 ^b	110 ± 6 ^c	<0.001
Uterine wt (mg)	43 ± 5 ^b	136 ± 13 ^c	93 ± 2 ^d	<0.001
Total uterine DNA (μg)	641 ± 66	833 ± 60	731 ± 32	NS
Total uterine RNA (μg)	231 ± 49	366 ± 54	235 ± 41	NS
Total uterine poly(A) RNA ^e	261 ± 63	609 ± 139	362 ± 45	NS
n ^f	5	8	4	

Data shown are the mean ± SEM. Values with different alphabetic superscripts are significantly different ($P < 0.05$, by SNK).

^a Based on one-way ANOVA.

^e Densitometric units.

^f Sample size.

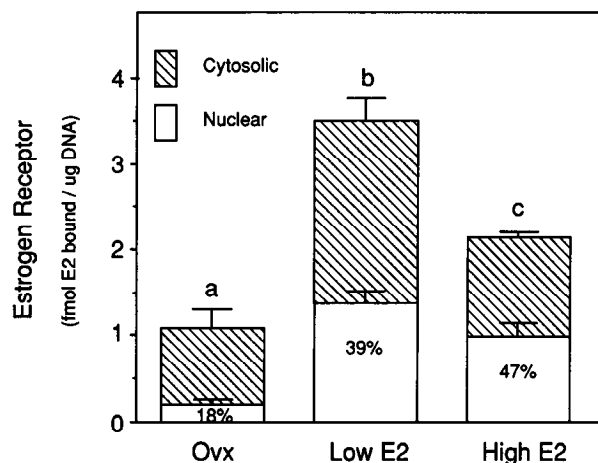


FIG. 4. Effect of estrogen treatment on uterine ER content and its intracellular distribution. Percentages designate the fraction of total ER bound tightly to chromatin and defined as nuclear binding. Note that cytosolic refers to presumably loosely bound nuclear ER that is found in the cytosol as a result of cell fractionation procedures. As shown by different letters, nuclear as well as cytosolic ER concentrations differed among all three treatment groups ($P < 0.05$, by SNK).

trus (*cf.* Table 1), the increases in E₂-exposed mice did not reach statistical significance.

Discussion

The results of this study reveal a strong association between E₂ and the biosynthesis and intracellular distribution of the uterine ER. Levels of plasma E₂ and uterine ER and ER mRNA were all elevated on proestrus and were at basal levels during the remainder of the cycle, except for E₂, which began to increase 1 day before proestrus. The increased levels of uterine ER and ER mRNA on proestrus were the consequence of increases in their intracellular concentrations and not increased numbers of ER-containing cells, since uterine DNA content did not change during the estrous cycle. The corresponding proestrous elevations of total and polyadeny-

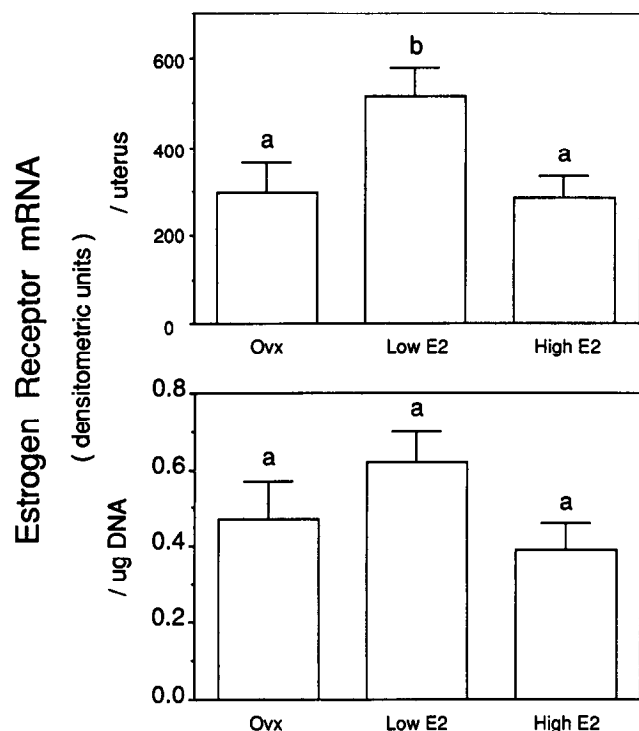


FIG. 5. Effect of estrogen treatment on uterine ER mRNA in OVX mice. Values with different letters differ significantly ($P < 0.05$, by SNK).

lated RNA revealed that the increase in ER mRNA was not the result of a selective activation of the gene encoding the ER, but was part of a more general change in uterine gene activity or turnover. E₂ administration at physiological doses produced changes in ER and ER mRNA in OVX mice similar to those observed in mice on proestrus, indicating that rising E₂ on proestrus probably underlies the increases in ER and ER mRNA on that day of the cycle.

The concentration of cell nuclear ER was markedly elevated on proestrus compared to that on metestrus, as reported previously in the rat (14) and hamster (15). It has generally been held that the increased nuclear binding of ER on proestrus reflects a shift of ER from the loosely bound nuclear compartment (*i.e.* cytosolic) to the tightly bound nuclear compartment as a consequence of increased binding of ER to estrogen and consequent activation of ER to a state with increased affinity for DNA. The present results are consistent with this view insofar as the proportion of ER that was activated increased on proestrus. However, increased activation of ER *per se*, as measured by the increase in the fraction of ER in the nuclear compartment, accounted for only 20% of the absolute increase in nuclear ER on proestrus. Indeed, most of the increase in nuclear ER was attributable to the increase in total ER protein. Total ER (*i.e.* that potentially available for nuclear binding) increased 1.8-fold on proestrus, whereas activated ER only in-

creased 1.3-fold. These results indicate that the up-regulation of total ER is the major factor accounting for the increased cell nuclear ER and consequent promotion of E₂-dependent biochemical processes in the uterus on proestrus. Numerous studies have shown that the regulation of gene transcription by estrogen and other steroid hormones is dependent on both nuclear occupancy (31, 32) and the total intracellular concentration of steroid hormone receptor (33–36), the latter by virtue of providing the pool of receptor available for nuclear binding.

Uterine ER mRNA increased 3-fold on proestrus, and this increase probably underlies the concomitant increase in ER protein. The increase in ER mRNA probably does not reflect a selective effect on ER transcription or turnover, however, since the total population of polyadenylated RNA increased on proestrus. It remains to be determined whether the increased ER mRNA reflects newly transcribed material or reduced turnover of extant message. However, the facts that there was a generalized increase in polyadenylated RNA and that E₂ increases RNA polymerase-II and the transcriptional rates of a number of genes (16) support the former possibility.

Both E₂ and progesterone modulate ER and ER action and, thus, could play a role in the increases in ER and its mRNA on proestrus. Progesterone generally antagonizes ER action in the uterus, by either reducing ER content or inhibiting nuclear occupancy of ER (29, 37, 38). Thus, it is unlikely that progesterone mediated the proestrous increase in ER or its message, unless the increase reflected a release from suppression by progesterone when it was at high concentrations earlier in the cycle (*i.e.* on proestrous evening or metestrous midday). However, if uterine ER were under negative control by progesterone during most of the cycle except on proestrus, one would expect long-term OVX mice to show increased ER, but this was not observed.

It is more likely that the preovulatory increase in E₂ underlies the proestrous increase in ER and its mRNA, since they were both increased by physiological levels of E₂ in OVX mice. To our knowledge, this is the first study to report that E₂ increases ER mRNA in the uterus. Earlier studies showed that E₂ decreases ER mRNA in both the rat uterus (17) and hypothalamus (18, 19), but levels of E₂ administered in both studies were supraphysiological. In the present study ER mRNA only increased in mice exposed to physiological concentrations of E₂; no increase occurred after exposure to higher concentrations of E₂. Similarly, total as well as nuclear ER levels were highest in mice exposed to physiological levels of E₂ and significantly lower in mice exposed to supraphysiological levels of E₂. Consistent with these observations, cytosolic rat uterine ER was slightly up-regulated at physiological concentrations of E₂, although it was down-regulated at

high E₂ concentrations (39). Also, in one report ER mRNA in MCF-7 breast cancer cells was slightly up-regulated at the lowest E₂ concentration (13), although, as in most other studies, it was down-regulated by higher concentrations of E₂ (10–12). Together, these observations indicate the importance of E₂ concentration to ER expression and indicate that ER and ER mRNA regulation as well as tissue responsiveness to E₂ may be biphasic. It is noteworthy that the uterine weight of mice exposed to supraphysiological levels of E₂ was markedly lower than that of mice exposed to physiological levels of E₂. Biphasic responses to E₂ have been reported previously. For example, low levels of E₂ increase the frequency of estrous cycles, whereas high concentrations suppress cycle frequency (23).

At present, there are no data to indicate how this biphasic regulation is accomplished. One possibility is that turnover and transcription of ER mRNA are differentially sensitive to E₂. At physiological concentrations, E₂ might promote transcription of ER mRNA and have little influence on ER turnover, whereas turnover might be enhanced at higher concentrations of E₂ to negate and ultimately overcome transcriptional enhancement.

In summary, the present study indicates that the basis for increased nuclear ER on proestrus involves two estrogen-dependent components: activation of ER to a nucleophilic state and increased concentration of ER available for activation, the latter accounting for most of the increase in nuclear ER. The process probably begins early on the day of proestrus with increased activation of ER in response to the rising levels of E₂. Increased activation of ER leads to increased DNA binding of ER, which increases ER mRNA and ER. Increased ER provides the substrate for a further increase in DNA-bound ER, which serves to amplify the effect of the rising E₂ in a feed-forward cascade that ultimately leads to the dramatic increases in ER mRNA, ER, and polyadenylated RNA that are observed at midday on proestrus. The other major observation of this study, that the increases in ER and ER mRNA are attenuated and sometimes suppressed when E₂ levels exceed physiological concentrations, raises intriguing questions about the underlying mechanisms and the biological and pharmacological significance of this biphasic response, and emphasizes the critical importance of hormone concentration in hormone action.

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